

## C PROOFS IN SECTION 6

In Section 6, we analyze the bounds of expected loss under four different optimization cases:

1. **Uniform Sampling + Learning Rate Schedule.**
2. **Ascending Data-Ordering + Practical WSD Schedule.**
3. **Ascending Data-Ordering + WSMD Schedule.**
4. **Ascending Data-Ordering + Stochastic Weight Averaging (SWA).**

In the following, we give the proof of their corresponding theoretical claims we mentioned in Section 6 one by one.

**Lemma C.1.** *Consider the uniform sampling, for any learning rate schedule  $E$  such that  $0 \leq \eta_i \leq 1$ , and the parameter initialized at  $(L, 0)$ , it holds that*

$$\min_E \bar{\mathcal{L}}(M; E) = \Omega(L^2).$$

*Proof.* We first consider the update rule of SGD in the optimization process on the x-axis as

$$w_t^{(1)} = w_{t-1}^{(1)} - \eta_t(w_{t-1}^{(1)} - x_t^{(1)}).$$

Then, taking the expectation over the randomness in SGD and the data generation gives

$$\begin{aligned} \mathbb{E}[w_t^{(1)}] &= (1 - \eta_t)\mathbb{E}[w_{t-1}^{(1)}] + \eta_t\mathbb{E}[x_t] \\ &= (1 - \eta_t)\mathbb{E}[w_{t-1}^{(1)}] + \eta_t \frac{M(M-1)d}{2} \\ &\geq \frac{M(M-1)d}{2}. \end{aligned}$$

Thus, we write out the lower bound for the expected loss

$$\mathbb{E}[\mathcal{L}(w_t)] = \mathbb{E}[w_t^\top w_t] \geq \mathbb{E}[w_t^{(1)} w_t^{(1)}] \geq \mathbb{E}[w_t^{(1)}] \mathbb{E}[w_t^{(1)}] = \Theta(L^2).$$

The above equation completes the proof of Equation (1).  $\square$

For Case 2 and Case 3, we give a more general lemma, for which the conclusions for Case 2 and Case 3 are direct corollaries.

**Lemma C.2.** *Consider the Ascending data-ordering, and a class of WSD learning rate schedules with the following formula*

$$\eta_t = \begin{cases} \eta_0 & \text{for } 1 \leq t \leq M - T_0 + 1 \\ \frac{1}{t - (M - T_0)} & \text{for } M - T_0 + 2 \leq t \leq M, \end{cases}$$

where  $T_0 = \omega(1)$ ,  $M - T_0 = \Theta(M)$  and  $\eta_0 = \frac{1}{2}$ , it holds for any learning rate schedule  $E$  with the above formula that

$$\bar{\mathcal{L}}(M; E) = \tilde{\Theta} \left( T_0^2 d^2 + \frac{L^2}{T_0} \right).$$

*Proof.* We write out the update rule on the x-axis in the Ascending data-ordering case

$$w_t^{(1)} = (1 - \eta_t)w_{t-1}^{(1)} + \eta_t x_1^{(M-t+1)}.$$

Using the above update rule, we can get the expression of  $w_M^{(1)}$  as

$$w_M^{(1)} = \prod_{t=M-T_0+2}^M (1 - \eta_t) w_{M-T_0+1}^{(1)} + \sum_{i=M-T_0+2}^M \prod_{j=i+1}^M (1 - \eta_j) \eta_i x_1^{(M-i+1)}$$

Then, plugging in the formula of the learning rate schedule gives

$$\begin{aligned}
w_M^{(1)} &= \prod_{t=M-T_0+2}^M \left(1 - \frac{1}{t - (M - T_0)}\right) w_{M-T_0+1}^{(1)} + \sum_{i=M-T_0+2}^M \prod_{j=i+1}^M (1 - \eta_j) \eta_i x_1^{(M-i+1)} \\
&= \prod_{k=2}^{T_0} \frac{k-1}{k} w_{M-T_0+1}^{(1)} + \sum_{i=M-T_0+2}^M \frac{1}{T_0} x_1^{(M-i+1)} \\
&= \frac{1}{T_0} w_{M-T_0+1}^{(1)} + \sum_{i=M-T_0+2}^M \frac{1}{T_0} x_1^{(M-i+1)}
\end{aligned}$$

The above equation uses the following fact

$$\begin{aligned}
&\left(1 - \frac{1}{T_0}\right) \cdot \left(1 - \frac{1}{T_0-1}\right) \cdots \left(1 - \frac{1}{i+1}\right) \cdot \frac{1}{i} \\
&= \frac{T_0-1}{T_0} \cdot \frac{T_0-2}{T_0-1} \cdots \frac{i}{i+1} \cdot \frac{1}{i} = \frac{1}{T_0},
\end{aligned}$$

where  $2 \leq i \leq T_0$ . Furthermore, we can find that

$$\begin{aligned}
w_{M-T_0+1}^{(1)} &= \prod_{t=1}^{M-T_0+1} (1 - \eta_t) w_0^{(1)} + \sum_{i=1}^{M-T_0+1} \prod_{j=i+1}^M (1 - \eta_j) \eta_i x_1^{(M-i+1)} \\
&= \frac{1}{2^{M-T_0+1}} w_0^{(1)} + \frac{1}{T_0} \sum_{i=1}^{M-T_0+1} \prod_{j=i+1}^{M-T_0+1} (1 - \eta_j) x_1^{(M-i+1)} \\
&= \frac{1}{2^{M-T_0+1}} w_0^{(1)} + \frac{1}{T_0} \sum_{i=1}^{M-T_0+1} \frac{1}{2^{M-T_0-i+1}} x_1^{(M-i+1)},
\end{aligned}$$

and then

$$\begin{aligned}
w_M^{(1)} &= \frac{1}{T_0} \sum_{i=M-T_0+2}^M x_1^{(M-i+1)} + \frac{1}{T_0} w_{M-T_0+1}^{(1)} \\
&= \frac{1}{T_0} \sum_{i=M-T_0+2}^M x_1^{(M-i+1)} + \frac{1}{T_0} \left( \frac{1}{2^{M-T_0+1}} w_0^{(1)} + \frac{1}{T_0} \sum_{i=1}^{M-T_0+1} \frac{1}{2^{M-T_0-i+1}} x_1^{(M-i+1)} \right) \\
&= \frac{1}{T_0} \frac{1}{2^{M-T_0+1}} w_0^{(1)} + \frac{1}{T_0} \sum_{i=1}^{T_0-1} x_1^{(i)} + \frac{1}{T_0^2} \sum_{i=T_0}^M \frac{1}{2^{i-T_0}} x_1^{(i)}.
\end{aligned}$$

Given that  $x_1^{(i)} = (i-1)d$ , and  $M - T_0 = \Theta(M)$ ,  $T_0 = \omega(1)$ , we can find that

$$\begin{aligned}
w_M^{(1)} &= \frac{1}{T_0} \frac{1}{2^{M-T_0+1}} w_0^{(1)} + \frac{1}{T_0} \sum_{i=1}^{T_0-1} x_1^{(i)} + \frac{1}{T_0^2} \sum_{i=T_0}^M \frac{1}{2^{i-T_0}} x_1^{(i)} \\
&= \frac{1}{T_0} \frac{Md}{2^{M-T_0+1}} + \frac{1}{T_0} \sum_{i=1}^{T_0-1} (i-1)d + \frac{1}{T_0^2} \sum_{i=T_0}^M \frac{1}{2^{i-T_0}} (i-1)d \\
&= o(d) + \frac{(T_0-1)(T_0-2)d}{2T_0} + \frac{d}{T_0^2} \sum_{i=0}^{M-T_0} \frac{i-1}{2^i} + \frac{T_0}{2^i} \\
&= o(d) + \Theta(T_0 d) + o(d) = \Theta(T_0 d)(1 + o(1))
\end{aligned}$$

Thus, the expected loss on the axis follows

$$\mathbb{E}[w_M^{(1)} w_M^{(1)}] = \Theta((T_0)^2 d^2).$$

Similarly, we write out the expected loss on the y-axis

$$\begin{aligned}\mathbb{E}[w_M^{(2)} w_M^{(2)}] &= \frac{1}{(T_0)^2} \sum_{i=1}^{T_0-1} \mathbb{E}[x_2^{(M-i+1)} x_2^{(M-i+1)}] (1 + o(1)) \\ &= \frac{L^2}{T_0} (1 + o(1)) \\ &= \Theta\left(\frac{L^2}{T_0}\right).\end{aligned}$$

The above equation completes the proof. Specifically, taking  $T_0 = M - \lfloor 0.9M \rfloor$  and  $T_0 = \Theta(M^{\frac{2}{3}})$  gives the results in Equation (2) and Equation (3).  $\square$

In the end, we show how a simple SWA method can beat the practical WSD schedule, which is stated in Theorem 6.1

*Proof of Theorem 6.1.* We first write out the expression for the parameter after a rescaled SWA as

$$\bar{\mathbf{w}}_M = \sum_{t=0}^{n-1} \frac{\alpha}{n} \mathbf{w}_{M-t}.$$

Then, plugging in the constant learning rate schedule gives

$$\bar{\mathbf{w}}_M = \sum_{i=1}^n \sum_{j=0}^{i-1} \frac{\alpha}{n} (1 - \eta_0)^j \mathbf{x}_i + \sum_{i=n+1}^M \sum_{j=0}^{n-1} \frac{\alpha}{n} (1 - \eta_0)^{j+i-n} \mathbf{x}_i + \sum_{i=1}^n (1 - \eta_0)^{M-i} \mathbf{w}_0.$$

We then decouple the parameter into the x-axis component and the y-axis component as  $\bar{\mathbf{w}}_M = (\bar{w}_M^{(1)}, \bar{w}_M^{(2)})$ . For the x-axis, we have  $\mathbf{x}_1^{(1)} = 0$ , thus we can rewrite the above equation in the x-axis

$$\begin{aligned}\bar{w}_M^{(1)} &= \frac{\alpha_0}{n} \eta_0 x_1^{(1)} + \eta_0 \left( \frac{\alpha_0}{n} (1 - \eta_0) + \alpha \right) x_1^{(2)} + \eta_0 \left( \frac{\alpha_0}{n} (1 - \eta_0)^2 + \alpha(1 - \eta_0) + \alpha \right) x_1^{(3)} \\ &\quad + \cdots + \left( \alpha_0 (1 - \eta_0)^{n-1} + \alpha \sum_{i=0}^{n-2} (1 - \eta_0)^i \right) x_1^{(n)} \\ &\quad + \sum_{i=n+1}^M \sum_{j=0}^{n-1} \frac{\alpha}{n} (1 - \eta_0)^{j+i-n} x_1^{(i)} \\ &\quad + \sum_{i=1}^n (1 - \eta_0)^{M-i} w_0^{(1)}.\end{aligned}$$

We then take  $\alpha_0 = \frac{1}{\eta_0}$  and  $\alpha = 1$ , then we recursively have

$$\begin{aligned}\frac{\alpha_0}{n} \eta_0 &= \frac{1}{n} \\ \left( \alpha_0 (1 - \eta_0)^j + \alpha \sum_{i=0}^{j-1} (1 - \eta_0)^i \right) &= \frac{1}{n} \quad \text{for all } 1 \leq j \leq n-1 \\ \sum_{j=0}^{n-1} \frac{\alpha}{n} (1 - \eta_0)^{j+i-n} &= \frac{1}{n} (1 - \eta_0)^{i-n} \quad \text{for all } 1 \leq j \leq n-1.\end{aligned}$$